II Jacobson Radical Theory

4 The Jacobson Radical

Exercise 1.1: secretly Lemma 4.1 part a)

Let R be a ring. Define the Jacobson radical $J(R)$ as the intersection of all maximal left ideals. Prove that $J(R)$ is, in fact, a two-sided ideal. (As a lemma, prove that an element $x \in R$ belongs to the Jacobson radical if and only if $1 - axb$ is invertible for all $a, b \in R$.)

Exercise 2.1: secretly Lemma 4.1 part b)

Let R be a ring and let $r \in R$. Prove that r belongs to $J(R)$ iff $rM = \{0\}$ for all simple left ${}_RM$.

Exercise 3.1: secretly Theorem 4.14 part a)

A ring is said to be **semiprimitive** if $J(R) = \{0\}$. Prove that if R is a semisimple ring, then R is artinian and semiprimitive.

Proof.

Suppose

$\blacksquare$

Exercise 4.1

Find $J(\mathbb{Z}_n)$ for all $n \in \mathbb{N}$.

Exercise 5.1

Let R, S be two rings. What is $J(R \times S)$?

Proof.

dupe of problem 8 below? ■

Exercise 6.1

Let R be a ring and let $n \geq 1$. What is $J(M_n(R))$?

Exercise 7.1

Let R be a ring with ideal I. What is $J(R/I)$?

Exercise 8.1: Exercise 4.12B

Let A, B be two rings. Show that $J(A \times B) = J(A) \times J(B)$.

Exercise 9.1: Exercise 4.21

Let $\mathrm{GL}_n(R)$ denote the group of units in the matrix ring $M_n(R)$. Suppose that I is an ideal of R contained in $J(R)$. Show that the map $\mathrm{GL}_n(R) \to \mathrm{GL}_n(R/I)$ is onto.

5 Jacobson Radical under Change of Rings

Exercise 10.1: Corollary 5.2

Let R be a commutative ring. Prove that the polynomial ring $R[x]$ is semiprimitive (meaning its Jacobson radical is zero) if and only if R is a reduced ring (meaning it has no nilpotents).

Exercise 11.1: Theorem 5.3

Prove that if k is a field and R a finitely-generated k-algebra, then R is semiprimitive.

Exercise 12.1: Proposition 5.6

Let R be a ring, and let A be a subring of R. Suppose that A is a direct summand of R (as a left R-module). Prove that $J(A) = J(R) \cap A$.

Exercise 13.1

Let R be a commutative ring. Prove that the intersection of all prime ideals of R is equal to the set of nilpotents in R.

Exercise 14.1

Let R be a commutative ring and let $p(x) = \sum_{i=0} a_i x^i$ be in $R[x]$. Prove that $p(x)$ is a unit in $R[x]$ if and only a_0 is a unit in R, and $a_1, \ldots, a_n$ are nilpotents in R.

Hint: show that $a_1, \ldots, a_n$ belong to every prime ideal of R. Thus use the previous exercise to deduce that $a_1, \ldots, a_n$ are nilpotent.

Exercise 15.1: Exercise 5.6

Let R be a ring, let $R[[t]]$ be the ring of formal power series in an indeterminate t with coefficients from R. Note that $tr = rt$ for all $r \in R$. Let (t) denote the ideal of $R[[t]]$ generated by t. This coincides with the set of power series with no constant term:

$$(t) := \{tf(t) : f(t) \in R[[t]]\} = \{f(t) \in R[[t]] : f(0) = 0\}$$

Prove that the Jacobson radical of $R[[t]]$ is:

$$J(R[[t]]) = J(R) + (t)$$

Exercise 16.1: secretly Theorem 4.14 part b)

Prove that if R is artinian and semiprimitive, then R is semisimple.

V Two Theorems

1 The Artin-Wedderburn Theorem

Exercise 1.1

Let R be a ring. Show that R is semisimple if and only if R is both artinian and semiprimitive.

Exercise 2.1: Chinese Remainder Theorem

Let R be a ring and let I, J be two ideals in R. There is a natural map $R \to (R/I) \times (R/J)$. Prove the Chinese Remainder Theorem: if $I + J = R$ (we say that I, J are **comaximal**), then this natural map is onto.

Exercise 3.1

Prove that every left primitive ring is prime, and that every semiprimitive ring is semiprime.

Exercise 4.1

Prove that if R is a commutative ring, then R is a domain iff R is prime. Give an example of a noncommutative prime ring which is not a domain.

Exercise 5.1

Let R be a ring and let I be an ideal of R. Prove that the following are equivalent:

i) I is a semiprime ideal of R, i.e. if $J^2 \subseteq I$ then $J \subseteq I$ for all ideals J.

ii) I is an intersection of prime ideals.

Exercise 6.1: Subdirect Products

Let $\{S_i\}_{i\in I}$ be a family of rings indexed by some (possibly infinite) set I, and let S denote their direct product:

$$S = \prod_{i\in I} S_i := \{(s_i)_{i\in I} : s_i \in S_i \quad \text{for all } i \in I\}$$

A subring R of S is said to be **full** if the projection $\pi_i|_R : R \to S_i$ is onto for all $i \in I$. A **subdirect product** of the S_i's is any ring which is isomorphic to a full subring of S.

Prove that R is a subdirect product of the S_i's if and only if there is a family of surjective ring homomorphisms $\varphi_i : R \to S_i$ such that $\bigcap_{i\in I} \ker(\varphi_i) = \{0\}$.

Exercise 6.2

Show that $\mathbb{Z}$ is a subdirect product of the family $\{\mathbb{Z}/n\mathbb{Z} : n \in \mathbb{N}\}$.

Exercise 6.3

Prove that a ring R is semiprime if and only if R is a subdirect product of prime rings.

The structure of primitive artinian rings

Let M be a semisimple right R-module, and let $\Delta := \mathrm{End}(M_R)$ be the ring of R-module endomorphisms of M. For these exercises, assume the Jacobson density theorem in the following form: *if $\varphi : {}_\Delta M \to {}_\Delta M$ is an endomorphism of M as a left Δ-module, and $x_1, \ldots, x_n \in M$ is a finite list of elements of M, then there exists $r \in R$ such that $\varphi(x_i) = x_i r$ for all $1 \leq i \leq n$.*

Exercise 7.1: The structure of primitive artinian rings

Our first goal is to find conditions under which the natural map $\Phi : R \to \mathrm{End}({}_\Delta M)$ is an isomorphism.

Prove that if R is artinian, then ${}_\Delta M$ is finitely generated.

Exercise 7.2

Use the Density Theorem to show that if ${}_\Delta M$ is finitely generated, then Φ is onto.

Exercise 7.3

Prove that if M_R is faithful, then Φ is one-to-one.

Exercise 7.4: Schur's Lemma

Our second goal is to interpret $\mathrm{End}({}_\Delta M)$ as a matrix ring over a division ring.

Prove that if M is a simple right R-module, then Δ is a division ring.

Exercise 7.5

Prove the general fact that if D is a division ring and N is a finitely generated left D-module, then $\mathrm{End}({}_D N)$ is isomorphic to the matrix ring $M_n(D)$.

You may freely use the fact that modules over division rings behave exactly like

vector spaces. That is: every module over a division ring has a basis, any two bases have the same cardinality, and linear mappings are determined by their action on a basis. Thus every finitely generated D-module is isomorphic to ${}_D D^n$ for some $n \geq 0$.

Exercise 7.6

Conclude that if R is a left primitive artinian ring, then R isomorphic to a matrix ring $M_n(\Delta)$, where $n \geq 1$ and Δ is a division ring.

Exercise 8.1

Prove that if R is artinian, then every primitive ideal of R is a maximal two-sided ideal.

Exercise 8.2

Let R be semisimple. Prove that R has at most finitely many primitive ideals.

Exercise 9.1: Artin-Wedderburn Theorem

Combine (almost) everything above to prove that if R is an artinian semiprimitive ring, then R is isomorphic to a finite direct product of matrix rings.

10 The Jacobson Commutativity Theorem

currently just going off of the discord msgs, actual exercises may change

Theorem 10.1: Jacobson Commutativity Theorem

Let R be a ring such that for each $a \in R$, there exists some number $n \in \mathbb{N}$ such that $a^n = a$. Then R is commutative.

Exercise 10.2

Prove it in the case that R is a division ring.

Exercise 10.3

Prove it in the case that $R = M_n(\Delta)$ is a matrix ring over some division ring Δ.

Exercise 10.4

Prove it in the case that R is semisimple. Hint: Use AWT.

Exercise 10.5

Prove it in the case that R is semiprimitive. Hint: Use JDT to reduce to when R is also artinian.

Exercise 10.6

Prove it for any ring R. Hint: By the previous exercise, $R/J(R)$ is commutative.